

DOCKER

- 1) Docker installed and version verification.

```
This message is shown once a day. To disable it please create the
/home/gauri/.hushlogin file.
[sudo] password for gauri:
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$ docker --version
Docker version 28.1.1, build 4eba377
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$
```

- 2) Running a test container

```
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:0e760fd9bc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
hello-world    latest    e2ac70e7319a   7 weeks ago   10.1kB
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS          PORTS          NAMES
a225d8618b97   hello-world    "/hello"             4 minutes ago   Exited (0) 4 minutes ago          friendly_sanderson
gauri@IN-1M96C54:/mnt/c/Users/gauridhadhiya$
```

- 3) Build docker file, tested image and run the image

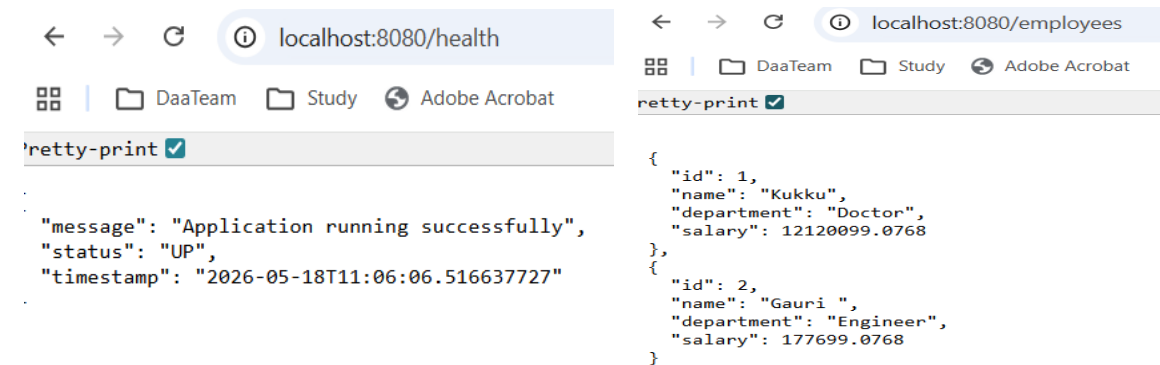
```
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker build -t springbootapp .
[+] Building 2.2s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 193B                             0.1s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine 1.1s
=> [internal] load .dockerignore                                 0.1s
=> => transferring context: 2B                                     0.1s
=> [1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine:sha256:5d14725f0e49e19df217f6ce179039f01ca25f5f0aa958573b467312599ca206 0.0s
=> [internal] load build context                                0.2s
=> => transferring context: 89B                                     0.2s
=> CACHED [2/3] WORKDIR /app                                    0.0s
=> [3/3] COPY target/SpringbootApp-0.0.1-SNAPSHOT.jar app.jar  0.2s
=> exporting to image                                           0.2s
=> => exporting layers                                           0.1s
=> => writing image sha256:59237ffa408a58df36cef908aa45b4f18133fc009fb2081522703eaa2295e2c 0.0s
=> => naming to docker.io/library/springbootapp                 0.0s
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
springbootapp latest    59237ffa408a   16 seconds ago 358MB
docker-springbootapp latest    fcef7fe15217   3 hours ago   394MB
hello-world    latest    e2ac70e7319a   7 weeks ago   10.1kB
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker run -p 8080:8080 springbootapp
```

4) Docker commands to list, start, stop, and remove containers.

```
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
436f933fd4df   springbootapp  "java -jar app.jar"     4 minutes ago  Exited (130) 19 seconds ago  PORTS          clever_varahamihira
97a10bc9996b   docker-springbootapp  "java -jar app.jar"     52 minutes ago  Exited (1) 51 minutes ago  PORTS          vigilant_tu
eea4c88bc056   docker-springbootapp  "java -jar app.jar"     3 hours ago    Exited (1) 3 hours ago    PORTS          goofy_lamarr
a225d8618b97   hello-world    "/hello"                6 hours ago    Exited (0) 6 hours ago    PORTS          friendly_sanderson
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker start 436f933fd4df
436f933fd4df
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
436f933fd4df   springbootapp  "java -jar app.jar"     5 minutes ago  Up 8 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  clever_varahamihira
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker stop 436f933fd4df
436f933fd4df
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
436f933fd4df   springbootapp  "java -jar app.jar"     6 minutes ago  Exited (143) 17 seconds ago  PORTS          clever_varahamihira
97a10bc9996b   docker-springbootapp  "java -jar app.jar"     53 minutes ago  Exited (1) 53 minutes ago  PORTS          vigilant_tu
eea4c88bc056   docker-springbootapp  "java -jar app.jar"     3 hours ago    Exited (1) 3 hours ago    PORTS          goofy_lamarr
a225d8618b97   hello-world    "/hello"                6 hours ago    Exited (0) 6 hours ago    PORTS          friendly_sanderson
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker rm 436f933fd4df
436f933fd4df
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
97a10bc9996b   docker-springbootapp  "java -jar app.jar"     53 minutes ago  Exited (1) 53 minutes ago  PORTS          vigilant_tu
eea4c88bc056   docker-springbootapp  "java -jar app.jar"     3 hours ago    Exited (1) 3 hours ago    PORTS          goofy_lamarr
a225d8618b97   hello-world    "/hello"                6 hours ago    Exited (0) 6 hours ago    PORTS          friendly_sanderson
gauri@IN-1M96C54:/mnt/c/Project/SpringbootApp$
```

5) Used Docker Compose to bring up the application and ensured all services are running correctly.

```
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$ docker compose up -d
WARN[0000] /mnt/c/Project/dockertestcrud/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Running 2/2
✓Container mysql-container      Started      1.3s
✓Container docker-springbootapp Started      2.7s
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$
```



localhost:8080/employees

pretty-print ☒

```
{
  "message": "Application running successfully",
  "status": "UP",
  "timestamp": "2026-05-18T11:06:06.516637727"
}
```

```
{
  "id": 1,
  "name": "Kukku",
  "department": "Doctor",
  "salary": 12120099.0768
},
{
  "id": 2,
  "name": "Gauri ",
  "department": "Engineer",
  "salary": 177699.0768
}
```

6) Tag Image for Docker Hub

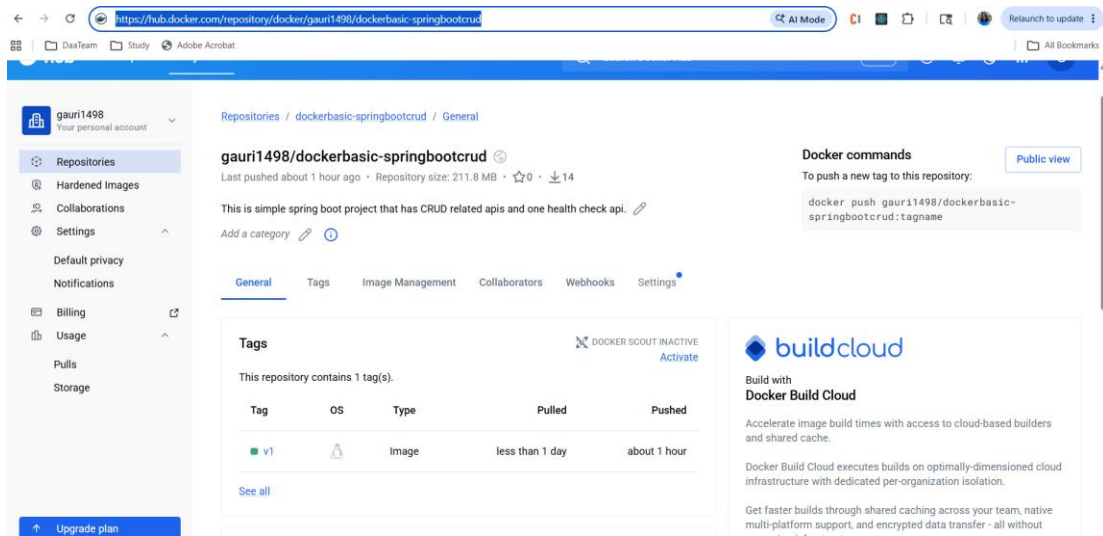
docker tag <local-image> <dockerhub-username>/<repo-name>:<tag>

eg: docker tag docker-springbootcrud gauri1498/dockerbasic-springbootcrud:v1

```
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$ docker tag dockertestcrud-app gauri1498/dockerbasic-springbootcrud:v1
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$
```

7) Push Image to Docker Hub

```
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$ docker tag dockertestcrud-app gauri1498/dockerbasic-springbootcrud:v1
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$ docker push gauri1498/dockerbasic-springbootcrud:v1
The push refers to repository [docker.io/gauri1498/dockerbasic-springbootcrud]
d433243c6e76: Pushed
aeb9a238e22: Pushed
9408fadd580f: Mounted from library/eclipse-temurin
5b58d0095bd1: Mounted from library/eclipse-temurin
d1890baff06e: Mounted from library/eclipse-temurin
430a1d12ca75: Mounted from library/eclipse-temurin
29df493baa13: Mounted from library/eclipse-temurin
v1: digest: sha256:6556508e955acb82004151c34179c5d95b29bd3d771bd71480ce39f66d1e7cec size: 1786
gauri@IN-1M96C54:/mnt/c/Project/dockertestcrud$
```



- 8) Pull the image from docker hub, check images and container, then go to that project path and run docker compose up -d

```

[ubuntu] password for gauri:
gauri@IN-1M96C54:/mnt/c/Users/gauridhadiya$ docker pull gauri1498/dockerbasic-springbootcrud:v1
v1: Pulling from gauri1498/dockerbasic-springbootcrud
Digest: sha256:6556508e955acb82004151c34179c5d95b29bd3d771bd71480ce39f66d1e7cec
Status: Image is up to date for gauri1498/dockerbasic-springbootcrud:v1
docker.io/gauri1498/dockerbasic-springbootcrud:v1
gauri@IN-1M96C54:/mnt/c/Users/gauridhadiya$ docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
gauri1498/dockerbasic-springbootcrud  v1          8aa55cbcd5a6   2 hours ago    394MB
dockertestcrud-app   latest      8aa55cbcd5a6   2 hours ago    394MB
docker-springbootapp latest      59237ffa40a8   19 hours ago   358MB
springbootapp        latest      59237ffa40a8   19 hours ago   358MB
mysql                 8          1487abffa4fa   5 days ago     812MB
hello-world           latest     e2ac78e7319a   7 weeks ago    10.1kB
gauri@IN-1M96C54:/mnt/c/Users/gauridhadiya$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
c9b2484709fa   springbootapp "java -jar app.jar"     18 hours ago   Up 18 hours   8080/tcp       condescending_wing
gauri@IN-1M96C54:/mnt/c/Users/gauridhadiya$ docker compose up -d
no configuration file provided: not found
gauri@IN-1M96C54:/mnt/c/Users/gauridhadiya$ cd ..
gauri@IN-1M96C54:/mnt/c/Users$ cd ..
gauri@IN-1M96C54:/mnt/c$ cd dockertestcrud
-bash: cd: dockertestcrud: No such file or directory
gauri@IN-1M96C54:/mnt/c$ cd project
gauri@IN-1M96C54:/mnt/c/project$ cd dockertestcrud
gauri@IN-1M96C54:/mnt/c/project/dockertestcrud$ docker compose up -d
WARN[0000] /mnt/c/project/dockertestcrud/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
[*] Running 3/3
✓Network dockertestcrud_default Created 1.5s
✓Container mysql-container Started 2.1s
✓Container docker-springbootapp Started 3.3s
gauri@IN-1M96C54:/mnt/c/project/dockertestcrud$

```